

- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Rare but serious skin condition known as drug reaction with eosinophilia and systemic symptoms (DRESS)
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Reported in a few patients, especially those with low BMIs, but not expected
- Less frequent and less severe than for most other antipsychotics

Sedation



- Some patients experience, especially at high doses

- May be less than for some antipsychotics, more than for others
- Usually transient and at higher doses
- Can be activating at low doses

What to Do About Side Effects

- Wait
- Wait
- Wait
- Usually dosed twice daily, so take more of the total daily dose at bedtime to help reduce daytime sedation
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- * For activating side effects at low doses, raise the dose
- * For sedating side effects at high doses, lower the dose
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- Schizophrenia: 40–200 mg/day (in divided doses) orally
- Bipolar disorder: 80–160 mg/day (in divided doses) orally
- 10–20 mg intramuscularly

Dosage Forms

- Capsule 20 mg, 40 mg, 60 mg, 80 mg
- Injection 20 mg/mL

How to Dose

- Schizophrenia (according to manufacturer): initial oral dose 20 mg twice a day; however, 40 mg twice a day or 60 mg twice a day may

be better tolerated in many patients (less activation); maximum approved dose 100 mg twice a day

- Bipolar disorder (according to manufacturer): initial oral dose 40 mg twice a day; on day 2 increase to 60 or 80 mg twice a day
- For intramuscular formulation, recommended dose is 10–20 mg given as required; doses of 10 mg may be administered every 2 hours; doses of 20 mg may be administered every 4 hours; maximum daily dose 40 mg intramuscularly; should not be administered for more than 3 consecutive days
- See also The Art of Switching section, after Pearls



Dosing Tips

*** More may be much more:** clinical practice suggests ziprasidone often under-dosed, then switched prior to adequate trials, perhaps due to unjustified fears of QTc prolongation

*** Dosing many patients at 20–40 mg twice a day is too low and in fact activating,** perhaps due to potent 5HT_{2C} antagonist properties

*** Paradoxically,** such activation is often reduced by increasing the dose to 60–80 mg twice a day, perhaps due to increasing amounts of dopamine 2 receptor antagonism

*** Best efficacy in schizophrenia and bipolar disorder is at doses >120 mg/day,** but only a minority of patients are adequately dosed in clinical practice

* Recommended to be taken with food because food can double bioavailability by increasing absorption and thus increasing plasma drug levels

- Meals of a few hundred calories (e.g., turkey sandwich and a piece of fruit) or more are necessary to enhance the absorption of ziprasidone
- Some patients respond better to doses >160 mg/day and up to 320 mg/day in 2 divided doses (i.e., 80–160 mg twice a day)
- Many patients do well with a single daily oral dose, usually at bedtime
- QTc prolongation at 320 mg/day not significantly greater than at 160 mg/day
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Ziprasidone intramuscular can be given short-term, both to initiate dosing with oral ziprasidone or another oral antipsychotic and to

treat breakthrough agitation in patients maintained on oral antipsychotics

- QTc prolongation of intramuscular ziprasidone is the same or less than with intramuscular haloperidol
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Rarely lethal in monotherapy overdose; sedation, slurred speech, transitory hypertension

Long-Term Use

- Approved to delay relapse in long-term treatment of schizophrenia
- Often used for long-term maintenance in bipolar disorder and various behavioral disorders
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- See Switching section of individual agents for how to stop ziprasidone

- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Mean half-life 6.6 hours
- Protein binding >99%
- Metabolized by CYP450 3A4
- Absorption is approximately doubled if taken with food



Drug Interactions

- Neither CYP450 3A4 nor CYP450 2D6 inhibitors significantly affect ziprasidone plasma levels
- Little potential to affect metabolism of drugs cleared by CYP450 enzymes
- May enhance the effects of antihypertensive drugs
- May antagonize levodopa, dopamine agonists
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval



Other Warnings/Precautions

- Ziprasidone prolongs QTc interval more than some other antipsychotics

- Ziprasidone is associated with a rare but serious skin condition known as drug reaction with eosinophilia and systemic symptoms (DRESS). DRESS may begin as a rash but can progress to other parts of the body and can include symptoms such as fever, swollen lymph nodes, swollen face, inflammation of organs, and an increase in white blood cells known as eosinophilia. In some cases, DRESS can lead to death. Clinicians prescribing ziprasidone should inform patients about the risk of DRESS; patients who develop a fever with rash and swollen lymph nodes or swollen face should seek medical care. Patients are not advised to stop their medication without consulting their prescribing clinician
- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Priapism has been reported
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and ziprasidone should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with

diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy

- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is taking agents capable of significantly prolonging QTc interval (e.g., pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin)
- If there is a history of QTc prolongation or cardiac arrhythmia, recent acute myocardial infarction, uncompensated heart failure
- If there is a proven allergy to ziprasidone

Special Populations

Renal Impairment

- No dose adjustment necessary
- Not removed by hemodialysis
- Intramuscular formulation should be used with caution

Hepatic Impairment

- No dose adjustment necessary

Cardiac Impairment

- Ziprasidone is contraindicated in patients with a known history of QTc prolongation, recent acute myocardial infarction, and uncompensated heart failure
- Should be used with caution in other cases of cardiac impairment because of risk of orthostatic hypotension

Elderly

- Some patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

- Not officially recommended for patients under age 18
- Clinical experience and early data suggest ziprasidone may be safe and effective for behavioral disturbances in children and adolescents

- Children and adolescents using ziprasidone may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Ziprasidone may be preferable to anticonvulsant mood stabilizers if treatment is required during pregnancy
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic>

Breast Feeding

- Unknown if ziprasidone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on ziprasidone should be monitored for possible adverse effects

The Art Of Psychopharmacology

Potential Advantages

- Some cases of psychosis and bipolar disorder refractory to treatment with other antipsychotics
- * Patients concerned about gaining weight and patients who are already obese or overweight
- * Patients with diabetes
- * Patients with dyslipidemia (especially elevated triglycerides)
- Patients requiring rapid relief of symptoms (intramuscular injection)
- Patients switching from intramuscular ziprasidone to an oral preparation

Potential Disadvantages

- Patients noncompliant with twice daily dosing
- * Patients noncompliant with dosing with food

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms

Pearls

- Head-to-head study in schizophrenia suggests lower metabolic side effects and comparable efficacy compared to some other atypical and conventional antipsychotics
- * When given to patients with obesity and dyslipidemia associated with prior treatment with another atypical antipsychotic, many experience weight loss and decrease in fasting triglycerides
- * QTc prolongation fears are often exaggerated and not justified since QTc prolongation with ziprasidone is not dose-related and few drugs have any potential to increase ziprasidone's plasma levels
- * Efficacy may be underestimated since ziprasidone is mostly underdosed (<120 mg/day) in clinical practice
- * Well accepted in clinical practice when wanting to avoid weight gain because less weight gain than most other atypical antipsychotics
- * May not have diabetes or dyslipidemia risk, but monitoring is still indicated

- Less sedation than some antipsychotics, more than others (at moderate to high doses)
- * More activating than some other antipsychotics at low doses
- Anecdotal reports of utility in treatment-resistant cases, especially when adequately dosed
- * A short-acting intramuscular dosage formulation is available
- Approved for mania in children ages 10–17
- Patients with inadequate responses to atypical antipsychotics may benefit from determination of plasma drug levels and, if low, a dosage increase even beyond the usual prescribing limits
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic

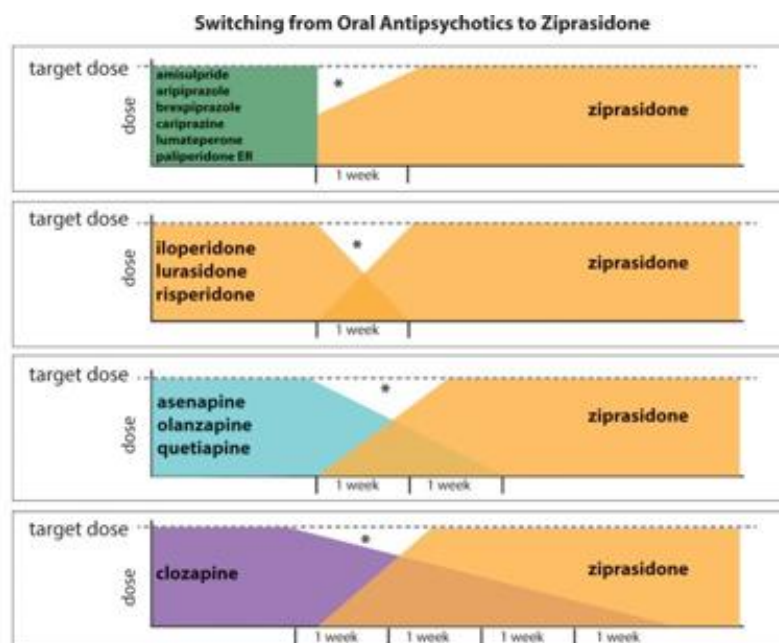
The Art Of Switching



Switching from Oral Antipsychotics to Ziprasidone

- With amisulpride, aripiprazole, brexpiprazole, cariprazine, lumateperone, and paliperidone ER, immediate stop is possible; begin ziprasidone at an intermediate dose

- Clinical experience has shown that quetiapine, olanzapine, and asenapine should be tapered off slowly over a period of 3–4 weeks, to allow patients to readapt to the withdrawal of blocking cholinergic, histaminergic, and alpha 1 receptors
 - Clozapine should always be tapered off slowly, over a period of 4 weeks or more
- * Benzodiazepine or anticholinergic medication can be administered during cross-titration to help alleviate side effects such as insomnia, agitation, and/or psychosis



Suggested Reading

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of

adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Komossa K , Rummel-Kluge C , Hunger H , et al. Ziprasidone versus other atypical antipsychotics for schizophrenia . *Cochrane Database Syst Rev* 2009 ;**7** (4):CD006627 .

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Nasrallah HA Jr . Atypical antipsychotic-induced metabolic side effects: insights from receptor-binding profiles . *Mol Psychiatry* 2008 ;**13** (1):27 – 35 .

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Taylor D . Ziprasidone in the management of schizophrenia: the QT interval issue in context . *CNS Drugs* 2003 ;**17** :423–30 .

Zolpidem

Therapeutics

Brands

- Ambien
- Ambien CR
- Intermezzo
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Non-benzodiazepine hypnotic; alpha 1 isoform selective agonist of GABA-A/benzodiazepine receptors

Commonly Prescribed for

(bold for FDA approved)

- **Short-term treatment of insomnia (controlled-release indication is not restricted to short-term)**

- **As needed for the treatment of insomnia when a middle-of-the-night awakening is followed by difficulty returning to sleep and there are at least 4 hours of bedtime remaining before the planned time of wakening (Intermezzo)**

How the Drug Works

- Binds selectively to a subtype of the benzodiazepine receptor, the alpha 1 isoform
- May enhance GABA inhibitory actions that provide sedative hypnotic effects more selectively than other actions of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects
- CR formulation may allow sufficient drug to persist at receptors to improve total sleep time and to prevent early morning awakenings that can be associated with the immediate-release formulation of zolpidem

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Improves quality of sleep

- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent



Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- Long-term adaptations of zolpidem immediate-release not well studied, but chronic studies of zolpidem CR and other alpha 1 selective non-benzodiazepine hypnotics suggest lack of notable tolerance or dependence developing over time

Notable Side Effects

- * Sedation
- * Dizziness, ataxia
- * Dose-dependent amnesia
- * Hyperexcitability, nervousness
- Rare hallucinations
- Diarrhea, nausea
- Headache



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with other CNS depressants in overdose
- Rare angioedema

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- To avoid problems with memory, only take zolpidem or zolpidem CR if planning to have a full night's sleep
- Lower the dose
- Switch to a shorter-acting sedative hypnotic
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- 10 mg/day at bedtime for 7–10 days (immediate-release)
- 12.5 mg/day at bedtime (controlled-release)

Dosage Forms

- Immediate-release tablet 5 mg, 10 mg
- Controlled-release tablet 6.25 mg, 12.5 mg
- Sublingual tablet 1.75 mg, 3.5 mg, 5 mg, 10 mg
- Oral spray 5 mg

How to Dose

- Men: 10 mg at bedtime for 7–10 days (immediate-release); 12.5 mg at bedtime for 7–10 days (controlled-release); 3.5 mg sublingually in the middle of the night if more than 4 hours of bedtime remain (Intermezzo)
- Women: 5 mg at bedtime for 7–10 days (immediate-release); 6.25 mg at bedtime for 7–10 days (controlled-release); 1.75 mg sublingually in the middle of the night if more than 4 hours of bedtime remain (Intermezzo)
- Intermezzo formulation is administered sublingually in the middle of the night; it should be placed under the tongue and allowed to dissolve completely before swallowing
- Intermezzo formulation should not be taken more than once per night



Dosing Tips

* Zolpidem is not absorbed as quickly if taken with food, which could reduce onset of action

- Patients with lower body weights may require only a 5 mg dose immediate-release or 6.25 mg controlled-release
 - Zolpidem should generally not be prescribed in quantities greater than a 1-month supply; however, zolpidem CR is not restricted to short-term use
 - Risk of dependence may increase with dose and duration of treatment
- * However, treatment with alpha 1 selective non-benzodiazepine hypnotics may cause less tolerance or dependence than benzodiazepine hypnotics
- Controlled-release tablets should be swallowed whole and should not be divided, crushed, or chewed

Overdose

- No fatalities reported with zolpidem monotherapy; sedation, ataxia, confusion, hypotension, respiratory depression, coma

Long-Term Use

- Original studies with zolpidem immediate-release did not assess long-term use
- Zolpidem CR is not restricted to short-term use
- Increased wakefulness during the latter part of the night (wearing off) or an increase in daytime anxiety (rebound) may occur with immediate-release and be less common with controlled-release

Habit Forming

- Zolpidem is a Schedule IV drug
- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

How to Stop

- Although rebound insomnia could occur, this effect has not generally been seen with therapeutic doses of zolpidem or zolpidem CR
- If taken for more than a few weeks, taper to reduce chances of withdrawal effects

Pharmacokinetics

- Short elimination half-life (approximately 2.5 hours)



Drug Interactions

- Increased depressive effects when taken with other CNS depressants
- Sertraline may increase plasma levels of zolpidem
- Rifampin may decrease plasma levels of zolpidem
- Ketoconazole may increase plasma levels of zolpidem
- Use with imipramine or chlorpromazine may be associated with decreased alertness

Other Warnings/Precautions

- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea
- Zolpidem and zolpidem CR should only be administered at bedtime
- Temporary memory loss may occur at doses above 10 mg/night
- Rare angioedema has occurred with sedative hypnotic use and could potentially cause fatal airway obstruction if it involves the throat, glottis, or larynx; thus if angioedema occurs treatment should be discontinued
- Sleep driving and other complex behaviors, such as eating and preparing food and making phone calls, have been reported in patients taking sedative hypnotics; in some cases these behaviors have resulted in serious injury or death, prompting the FDA to require a black box warning

Do Not Use